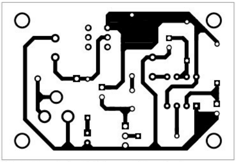

Fig. 4: Actual-size PCB layout of the vibration sensor

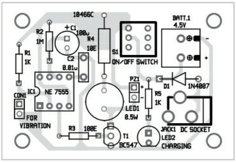

Fig. 5: Component layout of the PCB

timer is configured in monostable mode to activate the buzzer and the white LED for around two minutes when the sensor detects vibration. The vibration sensor is directly connected between trigger pin 2 and ground pin 1 of IC1. NE7555 is the CMOS version of NE555 timer and works off three volts.

The sensor is biased by resistor R1, which also keeps trigger pin 2 of IC1 in high state during standby. When the sensor senses a small vibration, its contacts close and takes pin 2 of timer to ground level. This triggers the timer and its output goes high for around two minutes based on the values of timing components R2 and C1. When output of the timer turns high, transistor T1 conducts to drive the 0.5W white LED and the buzzer.

The circuit is powered by a 4.5-volt rechargeable battery pack generally used in cordless phones. It can be charged using a mobile phone charger if a suitable socket is provided. LED2 indicates charging of the battery.

Construction and testing

An actual-size, single-side PCB layout for the vibration sensor is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB and enclose in a suitable box.

Connect the vibration sensor to the circuit using connector CON1. Glue the sensor on top of the box if it is to be used as a luggage protector, or on the window or door if used as a vibration alarm.

The circuit works off a 4.5V battery. The 5V regulated power supply is required to charge the battery. EFY

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